

Math 534H - Spring 2026 - Lecture 17

Now that we have the special solution $Q(x, t)$ with data $Q(x, 0) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$, how do

we use it to find the solution to the general ($k > 0$)

$$\text{IVP (H)} \begin{cases} u_t - k u_{xx} = 0 & x \in \mathbb{R}, t > 0 \\ u(x, 0) = \phi(x) & x \in \mathbb{R} \text{ for any} \end{cases}$$

$\phi: \mathbb{R} \rightarrow \mathbb{R}$ given.

Recall that derivatives of solutions are themselves solutions of the equation $u_t - k u_{xx} = 0$.

So : define

$$S(x,t) := \frac{\partial Q}{\partial x}(x,t)$$

Then

S also solves the heat equation $S_t - k S_{xx} = 0$
and so does :

$$u(x,t) := \int_{-\infty}^{\infty} S(x-y,t) \phi(y) dy, \quad t > 0$$

Define

Assume $\phi(y) \rightarrow 0$
 $|y| \rightarrow \infty$

Claim : u is the unique solution to the IVP (H)

Indeed let's check (2) the IC in (H) (last class)

$$u(x,t) = \int_{-\infty}^{\infty} \frac{\partial Q}{\partial x}(x-y,t) \phi(y) dy =$$

Prove! $= - \frac{\partial Q}{\partial y}(x-y,t)$

$$= - \int_{-\infty}^{\infty} \frac{\partial}{\partial y} [Q(x-y, t)] \phi(y) dy$$

$$\underline{\text{by IBP}} = \int_{-\infty}^{\infty} Q(x-y, t) \phi'(y) - Q(x-y, t) \phi(y) \Big|_{y=-\infty}^{y=+\infty}$$

Now plugging in $t=0$
and using that Q
has Heaviside IC
we have that

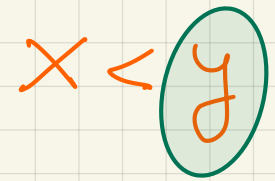
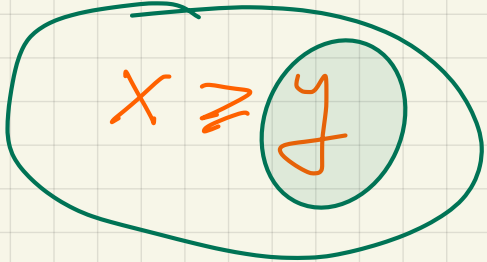


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Boundary terms = 0
are assumed to vanish
for example if ϕ vanishes
for $|y|$ large. But once we
explicitly understand $S(x-y, t)$ better
we won't need to assume
anything on ϕ to guarantee that
boundary terms vanish.

Recall $Q(x, 0) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$

$Q(x-y, 0) = \begin{cases} 1 & x \geq y \\ 0 & x < y \end{cases}$



$$\begin{aligned}
 u(x, 0) &= \int_{-\infty}^{\infty} Q(x-y, 0) \phi'(y) dy \\
 &\quad \uparrow \\
 &= \int_{-\infty}^x \phi'(y) dy = \phi(x) \quad \text{assuming} \\
 &\quad \lim_{|y| \rightarrow \infty} \phi(y) = 0.
 \end{aligned}$$

This gives (2) in the general IVP (H) (Last lecture)

We know u is a solution of $u_t - k u_{xx} = 0$ by the properties we saw last class and also know u is unique (use for ex. energy method for Heat eq.).

Finally, let us compute $S(x, t)$ explicitly

We need to compute $\frac{\partial Q}{\partial x}$; but by the FTC we get that

$$S(x, t) = \frac{\partial Q}{\partial x}(x, t) = \frac{1}{\sqrt{4\pi kt}} e^{-x^2/4kt}, \quad \underline{t > 0}$$

→ called Heat kernel, Heat propagator, or fundamental solution

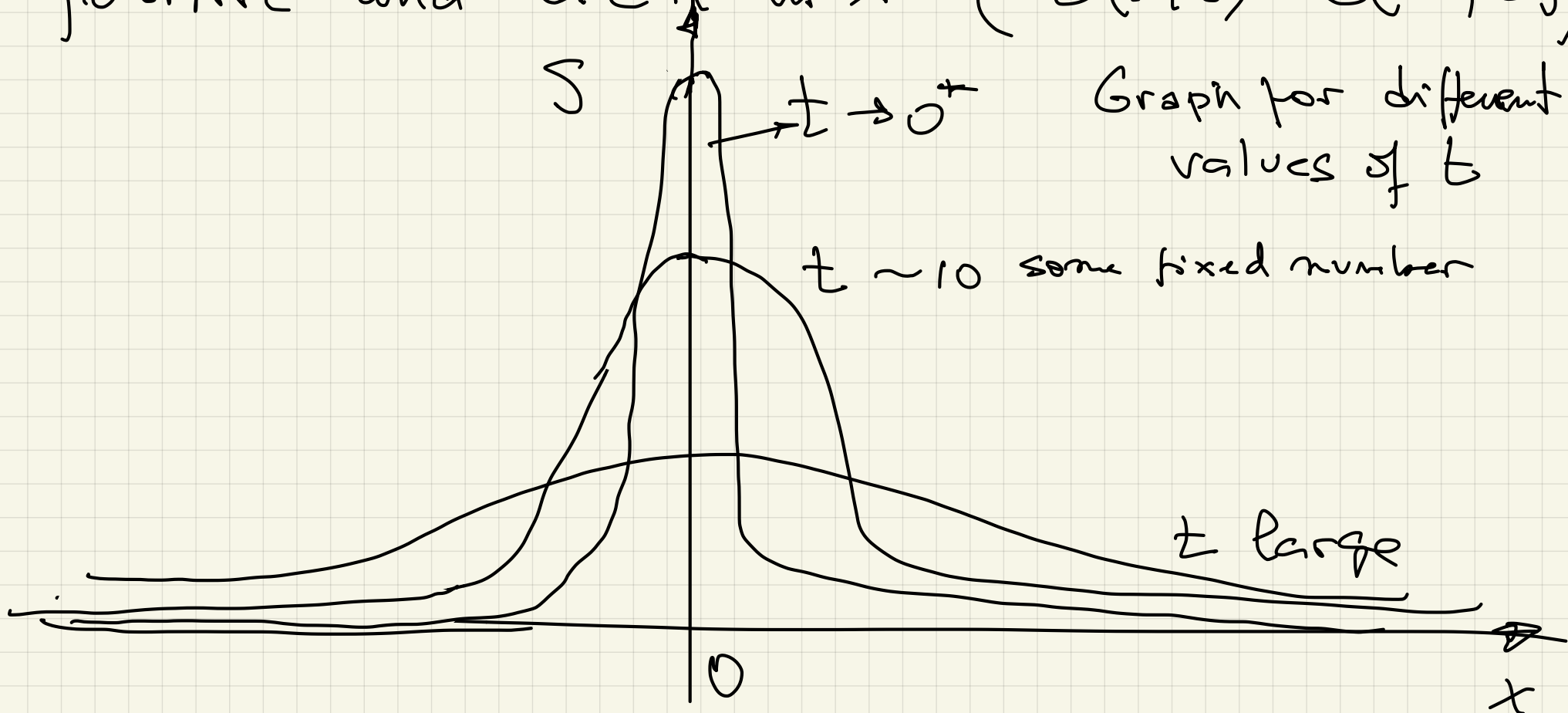
Hence

$$u(x, t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} e^{-(x-y)^2/4kt} \cdot \phi(y) dy$$

Properties of $S(x,t)$

1) $S(x,t) = \frac{1}{\sqrt{4\pi kt}} e^{-x^2/4kt}$ is

positive and even in x ($S(x,t) = S(-x,t)$)



For large t is a spread out Gaussian, for t intermediate is a Gaussian, but for $t \rightarrow 0^+$

$S(x, t)$ spikes at $x=0$ so that in the limit
you get "delta function"
(generalized function or distribution).
$$S(x) = \begin{cases} \infty & x=0 \\ 0 & x \neq 0 \end{cases}$$

The height of these spikes (max of $S(x, t)$ as $t \rightarrow 0^+$) is given by $\frac{1}{\sqrt{4\pi k t}}$.

2) Furthermore
$$\int_{-\infty}^{\infty} S(x, t) dx = 1 \quad \forall t > 0.$$

$$3) \max_{|x| \geq \varepsilon} S(x, t) \xrightarrow{t \rightarrow 0^+} 0 \quad \text{in fact}$$

($\varepsilon > 0$)

$$\max_{|x| \geq \varepsilon} S(x, t) = \frac{e^{-\varepsilon^2/4kt}}{\sqrt{4\pi kt}}$$

Coming back to $\hat{Q}(x, t)$, it is very hard to compute the integral. So one uses the following to express. Define (ERROR FUNCTION)

$$\text{Erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-p^2} dp$$

Then

$$Q(x, t) = \frac{1}{2} + \frac{1}{2} \operatorname{erf}\left(\frac{x}{\sqrt{4kt}}\right)$$

Homework (Class Hmwk)

Read ahead Example 2 in 2.4

pg. 51-52 before doing your homework.